

Dense Expands, Sparse Anchors: Channel-Asymmetric LLM Query Expansion for Hybrid Retrieval

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1 Introduction

LLM-based query expansion has become a common approach to improving retrieval effectiveness (Gao et al., 2023; Wang et al., 2023; Zhang et al., 2024). For hybrid retrieval, its gains are typically measured at a preset top- L cutoff using metrics such as nDCG and recall (Zhang et al., 2024; Liu and Zhang, 2025). Recent work on fixed top- L fusion shows that L does more than control access depth: changing the cutoff can also change which candidates and cross-channel contributions enter the fused ranking (Zhang, 2026). Consequently, a gain observed at one L reflects both the query expansion and the chosen cutoff; at another L , its magnitude—and even the ordering between methods—may change. Evidence from a single cutoff therefore does not isolate the effect of query expansion.

To remove this confound, we evaluate the retrieval effectiveness of each query construction using complete-list fusion, without truncating either channel (Zhang, 2026). We then reveal the dense and sparse rankings progressively and record the per-channel depths at which the same ordered top- K becomes fixed. This separates two questions: what fused ranking does an expansion produce, and how much ranked evidence is required to determine it? The answers need not move together. An expansion may improve retrieval effectiveness while pushing decisive evidence deeper into one channel. We therefore ask whether query expansion can improve effectiveness without increasing the access depth of either channel.

Meeting both goals requires treating dense and sparse retrieval differently. Prior studies show that expansion gains vary across retrievers, and that dense and sparse retrieval can favor different forms of generated evidence (Weller et al., 2024; Li et al., 2026). Our NFCorpus analysis shows what happens when one shared expansion is applied to both

channels. When the same generated passages are used in both channels, retrieval effectiveness improves and fewer dense ranks are needed to determine the fused result, yet more sparse ranks are required. The unexpanded sparse query matches only a small portion of the corpus; the added terms make far more documents eligible for sparse ranking, so the list no longer exhausts early. The same generated content therefore supplies useful semantic evidence to the dense retriever while spreading lexical evidence across a much larger sparse list. A shared expansion cannot control these two effects separately.

We address this asymmetry with channel-asymmetric query expansion. The LLM generates several complementary reference passages around the original query. For dense retrieval, orthogonal residual expansion preserves the original query direction and adds only the new semantic component supplied by the references. For sparse retrieval, score-product anchoring multiplies each document’s scores under the unexpanded and expanded queries. Documents matched only by generated terms receive no anchored score, while the new terms can still reorder documents supported by the original query. Together, these operations broaden semantic coverage without broadening sparse support: Dense expands; sparse anchors.

The evaluation mirrors this argument. A top- L sensitivity study first examines whether the measured gains and relative ranking of query expansion methods change with the cutoff. We then evaluate complete-list fusion on seven public retrieval benchmarks, comparing our method with the unexpanded query, shared expansion, and representative generative expansion methods in terms of retrieval effectiveness and per-channel access depth. Controlled experiments examine the contributions of complementary reference generation, orthogonal residual expansion, and score-product anchoring, followed by robustness analyses across generation

and retrieval settings. Together, these experiments test whether channel-asymmetric query expansion can produce better fused rankings while reducing the access required from both channels.

2 Related Work

2.1 LLM-Based Query Expansion

Short queries often omit expressions that appear in relevant documents. LLM-based query expansion supplies such expressions by generating text that resembles a relevant document. HyDE generates a hypothetical document and encodes it for dense retrieval (Gao et al., 2023), whereas Query2doc appends a generated pseudo-document to the original query for sparse and dense retrieval (Wang et al., 2023). Although their retrieval procedures differ, both methods use generated document-side language to enrich the information available from the original query.

Later work broadens the evidence produced by the LLM. Jagerman et al. elicit passages, keywords, and reasoning traces through different prompts (Jagerman et al., 2023), while MuGI samples multiple pseudo-references to cover alternative relevant expressions (Zhang et al., 2024). These approaches increase the variety of information available for retrieval, but the added content is not uniformly useful. Its effect varies across queries, datasets, and retrievers, and irrelevant generations may reduce retrieval effectiveness (Weller et al., 2024).

2.2 Integrating Expansions into Dense and Sparse Retrieval

Generated evidence can enter retrieval at different stages. Query2doc incorporates it into the query text (Wang et al., 2023), whereas MuGI additionally pools contextualized dense representations derived from multiple references (Zhang et al., 2024). Word2Passage reweights generated terms according to their reference granularity (Choi et al., 2025); Exp4Fuse combines rankings obtained from the original and expanded queries (Liu and Zhang, 2025); and DVCQR constructs separate lexical and semantic views for sparse and dense retrieval (Li et al., 2026). Existing methods therefore integrate generated evidence at the textual, representation, term-weighting, or ranking level.

Our method jointly designs reference generation and channel integration. It generates complementary passages that preserve the original query intent, then uses the same evidence asymmetrically: dense

retains its new semantic directions, whereas sparse uses its terms only to refine matches supported by the original query. Unlike multi-reference pooling or independently generated retrieval views, our construction couples intent-preserving generation with dense expansion and sparse anchoring.

2.3 Hybrid Fusion under Fixed Cutoffs

Query-expansion methods commonly evaluate retrieval effectiveness after taking a preset number of results from each ranking. MuGI follows this protocol for sparse, dense, and hybrid retrieval (Zhang et al., 2024), while Exp4Fuse combines fixed-depth rankings produced by the original and expanded queries (Liu and Zhang, 2025). In hybrid fusion, however, the cutoff determines both how much of each ranking is available and which cross-channel contributions enter the fused result. Reported expansion gains may therefore change with the selected depth.

EAHR separates the fused result from the depth needed to determine it by defining the target over complete rankings and progressively revealing each channel (Zhang, 2026). We use this distinction to evaluate query expansion. Each query construction is first assessed by the effectiveness of its complete-list fused ranking; dense and sparse access depths are then recorded when replay fixes the same ordered top- K . This evaluates both the ranking produced by an expansion and the concentration of evidence within its two channels.

3 Method

3.1 Method Overview

Given an original query q_o , the generator produces a set of complementary reference passages $R = \{r_1, \dots, r_n\}$ that preserve its intent while adding possible document-side expressions. The same references are then used to construct two channel-specific rankings. Orthogonal residual expansion adds their new semantic directions to the dense query representation. Score-product anchoring combines the original and expanded sparse scores for each document, allowing generated terms to reorder documents matched by the original query without expanding its lexical support. The resulting dense and sparse rankings are fused to produce the final result. Throughout this process, the original query provides the base and the generated passages provide incremental evidence: Dense expands; Sparse anchors.

3.2 Complementary Evidence Generation

Given q_o , the generator G produces n complementary reference passages:

$$R = G(q_o) = \{r_1, \dots, r_n\}.$$

Each passage is written as a plausible excerpt from a relevant document and introduces a different alias, technical term, or paraphrase. The prompt preserves the entities, relations, numbers, polarity, temporal conditions, scope, and constraints of q_o , while prohibiting broader information needs or invented answers. The passages therefore vary in expression while describing the same underlying query intent.

3.3 Channel-Asymmetric Query Construction

3.3.1 Orthogonal Residual Expansion

Let f_D denote the dense encoder. We first encode the original query as a unit vector

$$\mathbf{v}_o = \text{norm}(f_D(q_o)),$$

and average the representations obtained by concatenating q_o with each reference passage:

$$\bar{\mathbf{v}}_r = \frac{1}{n} \sum_{i=1}^n f_D(\text{concat}(q_o, r_i)).$$

We remove from $\bar{\mathbf{v}}_r$ the component parallel to $\mathbf{v}_o$, then add the remaining orthogonal residual to the original query:

$$\mathbf{e}_D = \text{norm}(\mathbf{v}_o + \bar{\mathbf{v}}_r - (\mathbf{v}_o^\top \bar{\mathbf{v}}_r) \mathbf{v}_o).$$

The original query remains the base direction, while the reference passages contribute only semantic directions not already represented by it. If the references provide no orthogonal component, then $\mathbf{e}_D = \mathbf{v}_o$. Documents are ranked by their similarity to $\mathbf{e}_D$, producing the dense ranking π_D .

3.3.2 Score-Product Anchoring

For sparse retrieval, we form an expanded query

$$q_r = \text{concat}(q_o, r_1, \dots, r_n).$$

A non-negative sparse scorer s_S evaluates each document d under the original and expanded queries:

$$s_o(d) = s_S(q_o, d), \quad s_r(d) = s_S(q_r, d).$$

We define the anchored score as their document-wise product:

$$s_A(d) = s_o(d) s_r(d).$$

A document receives a nonzero anchored score only if it matches both queries. Because q_r contains q_o , the anchored ranking retains the lexical support of the original query, while the generated terms change the relative scores within that support. Sorting documents by $s_A(d)$ produces the sparse ranking π_S .

3.4 Quality and Access Analysis

After constructing the dense ranking π_D and sparse ranking π_S , we fuse their complete rankings using weighted reciprocal rank fusion:

$$F(d) = \frac{w_D}{\kappa + \text{rank}_{\pi_D}(d)} + \frac{w_S}{\kappa + \text{rank}_{\pi_S}(d)}.$$

A missing rank contributes zero. Documents are ordered by $F(d)$, with document identifiers used to break ties, and the first K documents form the ordered result T_K . Retrieval effectiveness is evaluated on T_K .

We then replay incremental access to π_D and π_S . After reading L_c items from channel c , the largest contribution of its next unread item is

$$u_c(L_c) = \frac{w_c}{\kappa + L_c + 1}.$$

These bounds determine whether unread ranks can still change the members or order of the current top- K . If they can, the replay reads from the channel with the larger u_c , breaking ties in favor of Dense; otherwise, it stops and records (L_D, L_S) . The resulting measurements describe both the retrieval quality produced by each query construction and the depth at which its fused result becomes fixed.

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